

IRIN HOQUE ORCHI

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Research Interests

Natural Language Processing (NLP), Machine Learning, Mental Health Analysis, Low-Resource and Code-Mixed Languages, Social Media Text Analysis, Annotation Disagreement and Uncertainty-Aware Learning, Emotion and Sentiment Classification, Human-Centered AI.

Currently investigating: whether displaying model confidence scores changes how human annotators anchor on suggested labels, with a particular focus on high-disagreement items.

Education

BRAC University

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

December 2023

- GPA: 3.58/4.00 — Graduated with Distinction | IELTS: 7.0
- Relevant Coursework: Artificial Intelligence, Natural Language Processing, Data Structures & Algorithms, Database Systems, Software Engineering

Undergraduate Thesis

Mental Health Analysis of Cancer Patients with Lowest Survival Rates

2022–2023

Supervisor: Dr. Farig Yousuf Sadeque, Associate Professor, Dept. of CSE, BRAC University

- Constructed a novel dataset of 10,087 social media posts (Reddit, DailyStrength, HealthBoard) from cancer patients and caregivers across five cancer types selected for survival rates below 60%.
- Manually annotated 2,373 samples for emotional valence and arousal under multi-annotator agreement (Cohen's Kappa κ = 0.40–0.60, moderate); introduced a meta-reviewer protocol to resolve labeling conflicts rather than discarding disagreement as noise.
- Designed and compared three NLP pipelines for 4-class emotion classification: **GRU (TF-IDF + FastText embeddings)** — **75% accuracy, highest overall**; few-shot SBERT (all-MiniLM-L6-v2) — 71% accuracy, F1 = 0.82, precision = 0.81, recall = 0.83, competitive under a severely constrained annotation budget; baseline SVM (Bigram + TF-IDF) — 62% accuracy.
- Outperformed prior depression-detection baselines (Deshpande & Rao 2017: F1 = 0.80 on a binary dataset) despite a harder 4-class setting.

Publications

[1] Irin Hoque Orchi. “A Qualitative Study of the Mental Health Experiences of Cancer Patients with the Lowest Survival Rate and Their Caregivers.” Proc. 26th IEEE International Conference on Computer and Information Technology (ICCIT 2023), Cox's Bazar, Bangladesh, December 2023. **[First Author]**
DOI: 10.1109/ICCIT60459.2023.10441282

[2] Irin Hoque Orchi et al. “Mitigating Online Harassment: Machine Learning Approaches for Hate Speech Detection in Transliterated Bengali Comments (Pandulipi).” Proc. 26th IEEE International Conference on Computer and Information Technology (ICCIT 2023), Cox's Bazar, Bangladesh, December 2023. **[Co-Author]**
DOI: 10.1109/ICCIT60459.2023.10441244

Research Experience

- Pandulipi: Hate Speech Detection in Transliterated Bengali Comments (Co-Author)** — co-authored a study built on Pandulipi, the first publicly released transliterated Bengali hate speech corpus (5,000 Facebook comments; 2,837 hate / 2,163 non-hate), constructed by the project team, with 6 fine-grained categories (appearance, racial, sexual, religious, slang, other).
- Contributed to comparing 7 classifiers (SVM, Logistic Regression, Random Forest, Multinomial NB, Gaussian NB, Decision Tree, Bi-LSTM) across TF-IDF and Bag-of-Words representations.
- Best result — Binary classification: SVM with TF-IDF: accuracy = 0.743, F1 = 0.741; outperformed Bi-LSTM (accuracy = 0.576, F1 = 0.422), confirming that sparse feature methods dominate for short, noisy, code-mixed text.
- Best result — Multiclass classification: SVM: accuracy = 0.621, F1 = 0.578, compared to prior Hindi-English code-mixed baselines of 0.64–0.85 on larger, better-resourced corpora.
- Addressed the absence of standard tokenizers and stop-word lists for transliterated Bengali; proposed a custom preprocessing pipeline, since adopted as a baseline for follow-on work.

Research Dataset

Mental Health Insights of Vulnerable Cancer Patients

Kaggle | Mendeley Data

- 10,087 raw text samples from Reddit, DailyStrength, and HealthBoard; 2,373 samples manually annotated for emotional valence and arousal.
- Inter-annotator reliability: Cohen’s Kappa $\kappa = 0.40\text{--}0.60$ (moderate); disagreements resolved via meta-reviewer protocol.

Industry Experience

Software Engineer

August 2024 – Present

Bdjobs.com Limited

- Developing production-grade web applications (ASP.NET Core, Angular, RESTful APIs) following clean-architecture principles; led multiple features from design through deployment.
- Building an AI-based candidate-job matching score feature that ranks candidate profiles against job requirements.

Intern, IT Division

November 2023 – February 2024

Dhaka Bank PLC

- Implemented biometric and face-recognition technology for employee attendance systems.

Technical Skills

ML / NLP: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, SBERT (all-MiniLM-L6-v2), FastText, GRU/LSTM, few-shot learning

NLP Tasks: Text classification, emotion/sentiment analysis, hate-speech detection, TF-IDF, Bag-of-Words, word embeddings, corpus annotation, Cohen’s Kappa

LLMs & Uncertainty: LLM prompting, confidence-score extraction, uncertainty quantification, embedding-based similarity ranking

Languages: Python (primary), C++, TypeScript

Analysis: Pandas, NumPy, Jupyter Notebook, Google Colab, F1/precision/recall evaluation

Dev Tools: Angular, ASP.NET Core, Git, Firebase, Cloudflare Turnstile, LaTeX, PyCharm, VS Code, Visual Studio

Presentations & Outreach

- Presented paper [1] as first author at IEEE ICCIT 2023, Cox’s Bazar, Bangladesh (December 2023).
- Delivered a talk on the Digital Equity Fellowship 2023 at Save the Children Bangladesh, addressing digital equity, ICT skills, and gender inclusion in technology.
- Presented live demonstrations of web applications and projects developed during professional work when they went into production at Bdjobs.com Limited.

Extracurricular Activities

- Assistant Secretary, BRAC University Cultural Club (BUCUC) — 2020–2022
- Member, Photography Dept., BRAC University Adventure Club (BUAC) — 2020–2022
- Junior Executive, Finance & Marketing, Robotics Club of BRAC University (ROBU) — 2019–2022
- Volunteer, Kishor Alo, Prothom Alo — 2014–2018